

Graph-Based Drug Repurposing via Clinical Trial Mining and Network Analysis

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1. Introduction

1.1 Background on Drug Repurposing

Drug repurposing, also known as drug repositioning, refers to the identification of new therapeutic uses for existing or previously failed compounds. This approach offers a strategic advantage over de novo drug development by substantially reducing the time, cost, and risk associated with bringing a drug to market. Since repurposed drugs have often passed significant stages of clinical evaluation, their pharmacokinetic, toxicological, and manufacturing profiles are already well-established. This makes them strong candidates for treating diseases beyond their original indication, particularly in cases where unmet clinical needs or emerging global health threats demand rapid intervention.

While traditional repurposing efforts have relied on serendipity, clinician insight, or basic molecular similarity, recent advances in computational methods have enabled more systematic and data-driven approaches. Among these, network-based strategies have emerged as powerful tools for uncovering latent relationships between drugs and diseases, drawing on the complex interplay of biological, clinical, and phenotypic information.

1.2 Importance of Clinical Trial Data and Network-Based Approaches

Clinical trial registries, such as ClinicalTrials.gov, contain a wealth of curated and structured information on the real-world evaluation of pharmaceutical interventions. These datasets offer insight not only into a drug's intended use but also into the conditions under which it has been tested, the design of the study, and the outcomes measured. Importantly, many trials involve multi-arm comparisons, off-label explorations, or early signals of cross-indication efficacy, all of which are fertile ground for repurposing analysis.

By extracting and structuring this trial-level data, we can build bipartite networks that represent associations between drugs and diseases based on shared clinical co-occurrence. When projected into drug–drug or disease–disease spaces, these networks reveal topological features that can guide prioritisation for repurposing candidates. Centrality metrics, community structure, and connectivity patterns offer quantifiable indicators of a drug's potential relevance across multiple conditions.

In this work, we leverage clinical trial data from phases II to IV and construct a drug–disease bipartite network, which is then analysed using graph-theoretic techniques. We aim to provide a scalable and transparent framework that supports hypothesis generation for drug repurposing, grounded in real-world clinical evidence.

2. Clinical Trial Data Extraction

To construct a clinically grounded drug-disease network, we began by extracting trial-level data from ClinicalTrials.gov using its version 2 (v2) API. This RESTful interface allows programmatic access to structured metadata on thousands of registered interventional studies worldwide.

A Python-based client was implemented to query the database by therapeutic condition and retrieve up to 1,000 studies per term. Each returned study contained multiple protocol modules, from which relevant information such as condition(s), intervention(s), trial phase, and recruitment status were extracted and structured into JSON format.

Given the importance of ensuring medical and regulatory relevance, a strict eligibility filter was applied to all retrieved studies. Trials were retained only if they satisfied the following criteria:

- Status: Must be either *Recruiting*, *Active but not recruiting*, *Enrolling by invitation*, or *Completed*.
- Phase: Only trials listed as Phase II, III, or IV were considered, excluding early-phase or unspecified entries.
- Intervention Type: Only interventions classified explicitly as “Drug” were retained. Trials involving devices, behavioral therapies, or other modalities were excluded from further processing.

This initial filtering ensured that the resulting dataset reflected therapeutically meaningful and regulatory-stage evidence, reducing noise from exploratory or non-drug-related studies. The output formed the foundation for subsequent normalisation and network construction.

3. Entity Normalisation and Drug–Condition Pair Generation

Raw condition and intervention names extracted from clinical trials are highly heterogeneous. They may include varying capitalisation, synonyms, abbreviations, or additional qualifiers such as dosages and formulations. To facilitate accurate graph construction and analytical consistency, we implemented a robust entity normalisation pipeline grounded in the Medical Subject Headings (MeSH) vocabulary.

3.1 Mapping to MeSH Descriptors

We employed the 2025 MeSH descriptor XML file as a structured reference ontology for mapping. For each trial, all listed conditions and interventions were cross-referenced against MeSH terms. The ontology was parsed to extract canonical descriptors, each linked to a collection of entry terms, including synonyms and lexical variants.

Terms were categorised as either conditions or drugs based on their MeSH tree numbers, with disease-related entries originating from branches prefixed with “C” and pharmacological substances from branches prefixed with “D”. This allowed the classification of trial entries into medically meaningful domains.

3.2 Text Cleaning and Fuzzy Matching

To accommodate lexical inconsistencies, all input terms were preprocessed using the following transformations:

- Lowercasing
- Removal of parenthetical expressions and bracketed content
- Delimiting on conjunctions (e.g., “and”, “+”, “/”) and punctuation

Following preprocessing, a two-tiered matching approach was adopted. The system first attempted an exact match to MeSH terms; if none was found, it used a fuzzy matching heuristic based on string similarity (cut-off ≥ 0.85) to identify close alternatives. This helped resolve frequent real-world mismatches such as “paracetamol” vs. “acetaminophen” or “Type II Diabetes” vs. “Diabetes Mellitus, Type 2”.

3.3 Handling Unmatched Entries

Not all terms could be reliably resolved. Terms failing both exact and fuzzy matching were logged into a separate file for auditing. Reasons for mismatch were annotated (e.g., unmatched condition or unmatched intervention), and these entries were excluded from the downstream network. This deliberate curation step enhanced the interpretability and rigor of the final graph structure, avoiding the propagation of noise or misclassified nodes.

The resulting output was a structured list of matched (condition, drug) pairs, each optionally annotated with associated trial phases, ready for bipartite network construction.

4. Bipartite Network Construction

Following the generation of validated condition–drug pairs, we constructed an undirected bipartite graph to model the relationships between therapeutic agents and disease conditions as observed in clinical trials.

4.1 Building the Drug–Disease Bipartite Graph

Using the NetworkX library, we instantiated a bipartite graph in which nodes were partitioned into two distinct types:

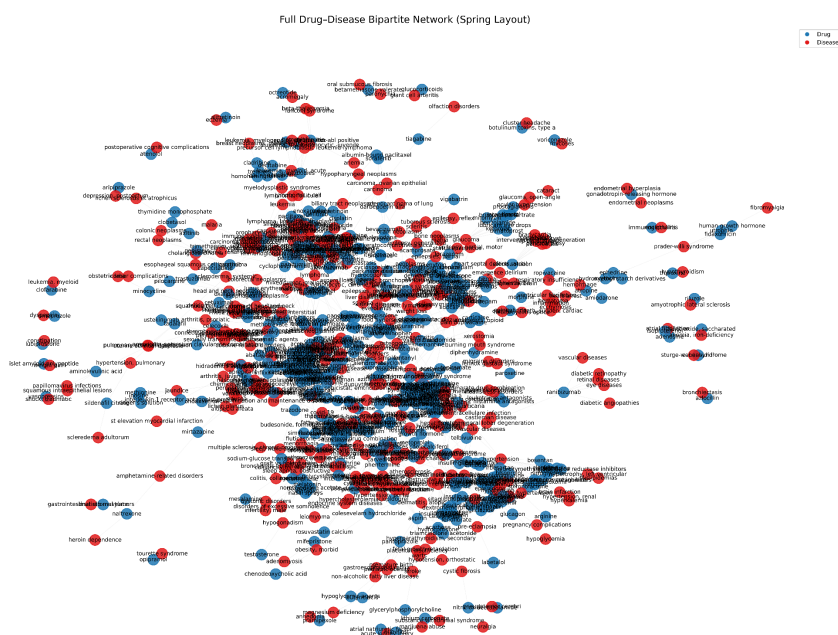
- Drug nodes, representing canonical intervention terms from MeSH
- Disease nodes, representing normalised clinical conditions

An edge was added between a drug and a disease if they co-occurred in at least one matched clinical trial, indicating that the drug had been evaluated for that condition. The graph structure thus encoded empirical evidence of drug–disease associations derived from real-world interventional studies.

4.2 Node Typing and Edge Logic

Each node was annotated with a "type" attribute ("drug" or "disease") to facilitate downstream filtering and projection. The bipartite nature of the graph was explicitly preserved, ensuring that edges only linked nodes of different types (i.e., no drug–drug or disease–disease edges at this stage).

The construction logic did not count edge multiplicity; repeated trials of the same drug for a disease contributed a single unweighted edge. This decision reflected our focus on structural connectivity and co-occurrence rather than trial frequency or dosage intensity.



4.3 Exclusion of Placebo Entries

To maintain clinical and pharmacological integrity, all intervention terms containing the word “placebo” were explicitly excluded from the graph. Placebos, although common in trial control arms, do not offer actionable repurposing insights and tend to introduce artificial hubs in the network due to their ubiquitous inclusion across trials. Skipping these entries ensured that the resulting graph reflected only biologically active interventions.

The final bipartite graph served as the substrate for drug network projection and centrality analysis, enabling the identification of drugs that span multiple disease nodes, a key indicator for repurposing potential.

5. Drug–Drug Network Projection and Centrality Computation

To identify candidate drugs with potential for repurposing, we transformed the bipartite graph into a one-mode drug–drug network. This projection enabled us to study direct relationships among drugs based on shared therapeutic targets.

5.1 One-Mode Projection of the Bipartite Graph

The bipartite graph was projected onto the drug node set using a weighted projection strategy. In this network, two drugs are connected by an edge if they have been evaluated for at least one common disease in the underlying clinical trial data. The weight of each edge corresponds to the number of shared disease nodes, reflecting the extent of therapeutic overlap between drugs.

This projected graph offers a new perspective: while the original bipartite network shows individual drug–disease associations, the projected network reveals clusters of pharmacological agents that target similar clinical areas, an indicator of potential functional similarity or repurposability.

5.2 Centrality Metrics for Prioritisation

To quantify the importance and influence of each drug within the projected network, we computed several standard graph-theoretic centrality metrics:

- Degree Centrality: Measures how many other drugs a given drug is connected to, indicative of therapeutic breadth.
- Eigenvector Centrality: Assigns higher scores to drugs connected to other highly connected drugs, highlighting central players in densely connected subspaces.
- Betweenness Centrality: Identifies drugs that serve as bridges between distinct clusters, useful for spotting repositioning candidates across disease families.
- Clustering Coefficient: Captures the tendency of a drug to form tightly knit groups with its neighbours.
- PageRank (Random Walk Centrality): Evaluates node influence based on the likelihood of reaching it via a random walk, balancing both popularity and network positioning.

These metrics provide complementary views of a drug's topological footprint within the repurposing landscape.

6. Results and Observations

The final bipartite graph and its projected drug–drug network revealed critical structural insights that inform the potential for drug repurposing across therapeutic areas.

6.1 Network Statistics

The bipartite graph constructed from phase II–IV clinical trials included:

- Drug nodes: 441
- Disease nodes: 353
- Edges: 1,200
- Placebo entries skipped: explicitly excluded during preprocessing

This resulted in a moderately dense network, capturing empirical co-occurrences of drugs and diseases as observed in interventional trials. The exclusion of placebos and unmatched terms ensured that the final graph was pharmacologically meaningful and methodologically sound.

6.2 Top Central Drugs

Centrality metrics calculated on the projected drug–drug network were used to identify drugs with the most prominent positions, an indicator of their potential relevance for repurposing. The results highlight distinct classes of drugs that dominate different parts of the therapeutic landscape.

Eigenvector centrality, which measures how well-connected a drug is to other influential drugs, ranked rosiglitazone as the most structurally dominant compound in the network. It exhibited the highest score (0.1693), followed closely by ritonavir (0.1584), rifampin (0.1568), leucovorin (0.1505), and alendronate (0.1473). These drugs occupy highly central positions in tightly knit therapeutic communities and are often evaluated alongside other widely tested agents.

In terms of PageRank, which reflects the probability of reaching a drug through a random walk across the network, cyclophosphamide emerged as the top-ranked node (0.0082), suggesting its broad reach and integration across the network. It was followed by dexamethasone (0.0068), metformin (0.0057), saline solution (0.0045), and epinephrine (0.0045). These agents are not only common across diverse trials but also strategically positioned to influence multiple disease areas.

For degree centrality, which simply counts how many distinct diseases a drug connects to, rosiglitazone once again led with a strikingly high value of 0.3205, meaning it was linked to over 32% of all disease nodes. Other high-degree drugs included leucovorin (0.2432), atorvastatin (0.2273), prednisone (0.2182), and ritonavir (0.2182). These drugs form the backbone of polytherapeutic trial design and are tested across broad clinical domains.

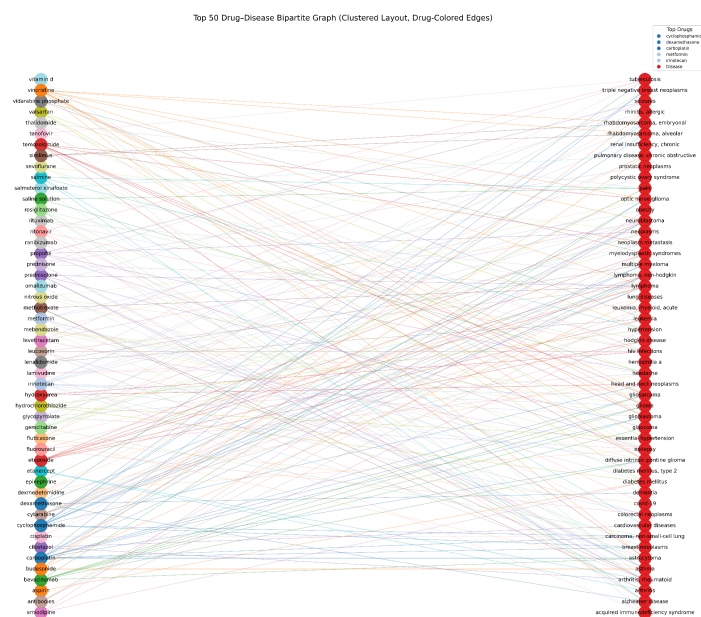
Lastly, betweenness centrality, which captures how frequently a drug acts as a bridge between otherwise unconnected disease clusters, further underscored the pivotal role of rosiglitazone (0.1141). Other notable bridging agents were cyclophosphamide (0.0614), prednisolone (0.0585), metformin (0.0534), and prednisone (0.0532), all of which likely contribute to cross-domain treatment strategies or signal repositioning opportunities into novel therapeutic spaces.

What is particularly notable is that rosiglitazone consistently appears at the top across eigenvector, degree, and betweenness centrality rankings. Its structural prominence in the network, combined with its broad disease coverage, positions it as a highly attractive candidate for repurposing. Its central location within the therapeutic landscape means it may either have unexplored indications or warrant reevaluation in the context of modern precision medicine strategies.

6.3 Observed Clustering of Drug–Disease Relationships

Visualisation of the top-50 subgraph (clustered layout) revealed structured clusters where groups of drugs co-occurred with overlapping disease sets. This was particularly evident among anti-inflammatory and oncology-related compounds, forming tightly knit subgraphs. Drugs like prednisone and dexamethasone appeared in multiple such clusters, underscoring their broad clinical footprint.

In contrast, the presence of drugs with high betweenness centrality, such as cyclophosphamide and metformin, points to their potential in connecting therapeutically distinct disease spaces, offering opportunities for strategic repositioning beyond their traditional domains.



7. Discussion

7.1 Interpretation of Highly Central Drugs

The network centrality analyses revealed a consistent set of drugs occupying structurally dominant positions within the projected drug–drug network. Rosiglitazone, in particular, emerged as the most influential agent across multiple dimensions, degree, eigenvector, and betweenness centrality, suggesting a broad clinical footprint and a bridging role across disparate disease groups. Its historical use in metabolic disorders, combined with observed connectivity to conditions outside this domain, supports the hypothesis that it may have latent therapeutic potential beyond its current indications.

Similarly, drugs such as cyclophosphamide, metformin, and dexamethasone surfaced as multi-indication agents with high PageRank and betweenness scores. These compounds are frequently deployed in systemic or combinatorial therapies, which may explain their broad reach across disease clusters. Their network prominence indicates not only historical trial popularity but also their likely relevance as anchor therapies in future repositioning strategies.

7.2 Potential Implications for Repurposing

Highly central drugs in the network can be prioritised for repurposing efforts on several grounds. First, their connectivity implies they have been clinically tested across diverse pathologies, reducing the regulatory and safety uncertainty often associated with novel agents. Second, their appearance in multiple disease subgraphs hints at mechanistic versatility or immunomodulatory effects, both of which are desirable properties in repositioning candidates.

From a drug development perspective, these agents represent low-hanging fruit for adaptive trials, off-label exploration, or AI-guided hypothesis generation. The graph-based approach used here can be extended to assess similarity across drug clusters, identify underexplored edges, or simulate perturbations that mimic intervention redirection.

7.3 Limitations

While the current framework offers a data-driven route to repurposing, several limitations should be acknowledged:

- Dependence on public clinical trial data: The analysis is confined to studies registered on ClinicalTrials.gov, potentially excluding regional trials or unpublished data. As a result, certain drug–disease pairs may be underrepresented or absent.
- MeSH mapping accuracy: Although robust matching logic was employed, including exact and fuzzy techniques, the process is inherently sensitive to term ambiguity, synonym resolution, and the coverage limitations of the MeSH ontology. Some conditions or interventions may not have mapped cleanly, leading to loss of valid pairs or misclassification.

- **Static representation:** The network captures associations at the time of data retrieval but does not encode temporal trends, trial outcomes, dosage information, or adverse effects, all of which are relevant for robust repurposing decisions.

Despite these limitations, the results underscore the potential of graph-theoretic approaches to uncover repurposing opportunities rooted in clinical reality. Integrating this network foundation with mechanistic data, literature mining, or Bayesian modelling can further enhance its predictive power and translational value.

8. Conclusion and Future Work

8.1 Summary of Findings

This study demonstrates the feasibility and power of leveraging clinical trial data to construct a drug–disease bipartite network as a foundation for drug repurposing. By systematically extracting, normalising, and structuring trial data from ClinicalTrials.gov, we built a robust network capturing over 1,200 empirically validated associations between 441 drugs and 353 diseases. Projecting this network into a drug–drug space allowed for the identification of structurally central compounds, most notably *rosiglitazone*, *cyclophosphamide*, and *metformin*, whose prominence suggests broad therapeutic relevance and high potential for repositioning.

Centrality analyses provided multiple, converging signals of influence and connectivity, revealing drug hubs that act as bridges across disease domains. These results offer strong, data-backed hypotheses for repurposing and support a scalable graph-based framework for prioritising candidates.

8.2 Integration with Bayesian Models and LLMs

While the network structure offers a static, topological view of potential repurposing opportunities, its predictive utility can be significantly enhanced by incorporating probabilistic reasoning and semantic intelligence. In the next stage of this work, we aim to integrate

- **Bayesian models** to assign probabilistic confidence scores to drug–disease pairs, incorporating priors from graph features, literature frequency, or known mechanisms of action.
- **Large Language Models (LLMs)** to reason over biomedical literature and trial descriptions, identify unstructured signals of efficacy or safety, and generate interpretable narratives explaining why a particular drug might be viable for a new indication.

This multi-layered approach, combining structured graphs, statistical inference, and unstructured language understanding, can move beyond co-occurrence and into mechanistically and clinically informed repurposing.

8.3 Potential for Downstream Validation and Clinical Translation

The framework outlined here is well-positioned for downstream applications. High-centrality candidates identified via this network can be validated against biological pathways, preclinical data, or real-world evidence. Further, the graph structure lends itself to simulation of intervention effects, exploration of drug–drug interactions, and targeted experimental design for adaptive or platform trials.

With proper integration into translational pipelines, this system could assist academic researchers, regulatory agencies, and pharmaceutical companies in narrowing the search space for novel indications, accelerating the time to impact while reducing development cost and attrition.